

RISHAB RAJ

Bachelor of Technology, Civil Engineering — National Institute of Technology, Raipur

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EDUCATION

National Institute of Technology, Raipur <i>B.Tech in Civil Engineering</i>	Aug. 2024 - Present
Shanti Niketan Academy, Gaya, Bihar <i>Central Board of Secondary Education (Class XII)</i>	2022 - 2023
D.A.V. Public School, Cantt. Area, Gaya, Bihar <i>Central Board of Secondary Education (Class X)</i>	2020 - 2021

EXPERIENCE

Indian National Science Academy (INSA) Research Fellow — IIT Bombay <i>INSA Fellow at IIT Bombay</i>	May 2026 - Jul 2026
- Designed a hybrid Population Synthesis pipeline by integrating Variational Autoencoders (VAE) with Iterative Proportional Updating (IPU) to generate realistic census-scale synthetic populations from limited survey data. - Generated diverse synthetic population profiles using a VAE and recalibrated them with IPU to preserve demographic relationships while ensuring consistency with official census statistics for agent-based transportation microsimulation.	

PERSONAL PROJECTS

Embrya — Embryo Viability Analysis Platform <i>Built a full-stack clinical-decision-support platform that scores IVF embryo viability from microscope images.</i>	Dec 2025
- Designed an ensemble machine learning pipeline of three SMOTE-balanced Random Forest classifiers to predict embryo viability from extracted morphological features. - Built an OpenCV-based feature extraction engine computing circularity, symmetry, fragmentation, edge density, and entropy from embryo images for model inference. - Engineered a FastAPI backend exposing a REST prediction endpoint with request validation, feature scaling, and probability-averaged ensemble inference. - Developed a React + TypeScript clinical dashboard (Tailwind CSS, Radix UI) with real-time scoring, batch processing, embryo comparison, and PDF/TXT report generation.	
IR Image Colorization & Enhancement <i>Satellite infrared-to-RGB translation system using super-resolution, semantic segmentation, and conditional image synthesis.</i>	Jul 2026
- Built a 4-stage pipeline (Real-ESRGAN → DeepLabV3+ → Pix2PixHD → Semantic Correction) to colorize Landsat 8/9 infrared satellite tiles into realistic RGB imagery, trained on 4,940 paired IR/RGB tiles from USGS. - Trained a DeepLabV3+ segmentation model (ResNet-101 backbone, adapted to single-channel IR input) using auto-generated pseudo-labels via spectral indices (ExG, NDWI), eliminating the need for manual annotation. - Designed a 4-channel conditioned Pix2PixHD generator (IR + one-hot segmentation map) with a multi-scale PatchGAN discriminator, achieving semantically consistent colorization with explicit land-cover class awareness. - Deployed an end-to-end inference dashboard (Gradio) enabling upload of a raw IR image and real-time output of the full enhance → segment → colorize pipeline result.	

TECHNICAL SKILLS AND INTERESTS

Programming Languages: Python, JavaScript, HTML, CSS

Data Analytics: Data Analysis, Microsoft Excel, SQL, NumPy, Pandas

Machine Learning: Machine Learning

Developer Tools: VS Code, Jupyter Notebook

Soft Skills: Analytical Thinking, Problem Solving, Process Optimization, Teamwork, Leadership, Communication

Areas of Interest: Artificial Intelligence, Data Analytics, Transportation Engineering

POSITIONS OF RESPONSIBILITY

Executive, Civil Engineering Association (CEA), NIT Raipur	Jul 2025 - Present
Coordinator, National Service Scheme (NSS), NIT Raipur	Oct 2024 - Present

ACHIEVEMENTS

Amazon ML Summer School 2026 Scholar, Amazon	Jul 2026
INSA Summer Research Fellow, Indian National Science Academy (INSA)	May 2026
National Finalist, Smart India Hackathon (SIH)	Dec 2025
Selected Intern, NHA Winter Paid Internship	Dec 2025
Runner-up, Truss Bridge Making Competition, Civil Engineering Department, NIT Raipur	Feb 2025
Runner-up, Mad Over Marketing — Business Case Study Competition, E-Cell NIT Raipur	Mar 2025
National Level Merit Holder, Indian Olympiad Qualifier in Mathematics (IOQM)	Sept 2020
National Level Merit Holder, AIR (15), Indian National Mathematics Talent Contest	Nov 2018